

Machine Learning Introduction

1. Introduction to machine learning (Completed)
2. AI vs ML vs DL vs DS (Completed)
3. Types of ML (completed)
4. Train Test Split

Introduction to machine learning :-

Some of the examples of the machine learning:-

1. Google photos
2. Email Spam ham detection
3. Autonomous Car
4. Gboard Autocorrector
5. Google Suggestions
6. Product Recommendation in Ecommerce Sites
7. Instagram Recommendation (Reels, Posts)

Q. What is machine learning?

Machine Learning is the ability of the learning of the machines by itself.

or

We can say that it is the subfield of the computer science that gives the ability to learn without explicitly programmed.

or

A computer program is said to learn from the experience E with respect to some task and the performance measure P .

Q. Why ML is famous in now a days?

- > Processors are very strong for now a days.
- > Data is new oil...
- > Huge data

AI vs ML vs DL vs DS

AI -> Artificial Intelligence is the branch of computer science in which it mimics the human brain..

Ex - Tesla (Autonomous car)

ML -> Sub branch of ai

DL -> Deep Learning (Neural Networks)

Example -> Image Recognition , Object Detection (CNN), RNN(Siri)

Data Science -> Data science is the intersection of all the above technologies..

Data Collect -> Analyse (EDA) -> Recommendation System

Types of ML -

1. Supervised Learning
2. Unsupervised Learning
3. Semi-Supervised Learning
4. Reinforcement Learning

1. Supervised learning :-

Teacher → 20 questions, Solutions → Question? →

Supervised learning is the type of machine learning in which we work on the labelled data

for example → House price prediction

feature1(area), f2(location), f3, Price (Y) → Labelled data

Stock Price Prediction →

f1, f2, f2 | Price (Labelled data)

Example -

1. Next word Predictor (Here the label is the next word which we want as suggestion)
2. Music Recommendation system (

Unsupervised Learning :-

It is the type of machine learning in which the data is given but it is not labelled, but on the hidden pattern it cluster or divide into the different categories..

cluster 1 → red, round (Seb)

cluster 2 → yellow, long, (Banana)

cluster 3 -> yellowish, oval (Mango)

Example - Google Photos, Customer Segmentation

the people who buys something -> cluster salary -> max -> 5000 15% discount.

Semi Supervised ->

Example ->

Google Photos -> unsupervised learning(Clusters the different groups of photos) -> Supervised Learning

Netflix -> Recommendation of movies (Supervised)

It divides the profile on the basis of the some of features like (Genre, Language)

Reinforcement Learning -> Learning from the environment.

Example ->

what is the machine learning ?

ans -> thumps

autonomous car -> +10 , accident -100 reward system

data set -> backbone

Train test split ->

1. Train Data
2. Validation data
3. Test data

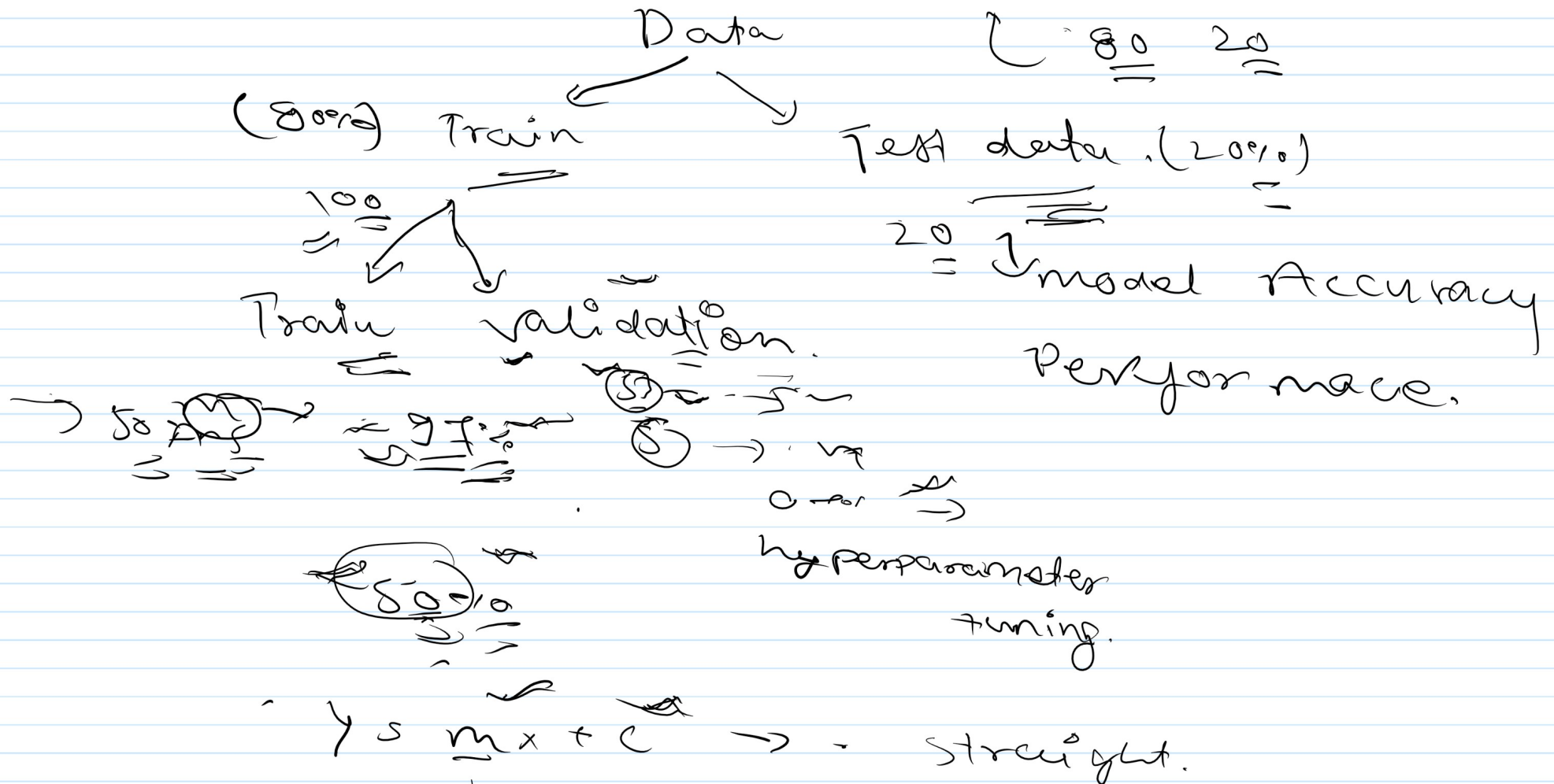
Train data -> From this data we basically train the model.

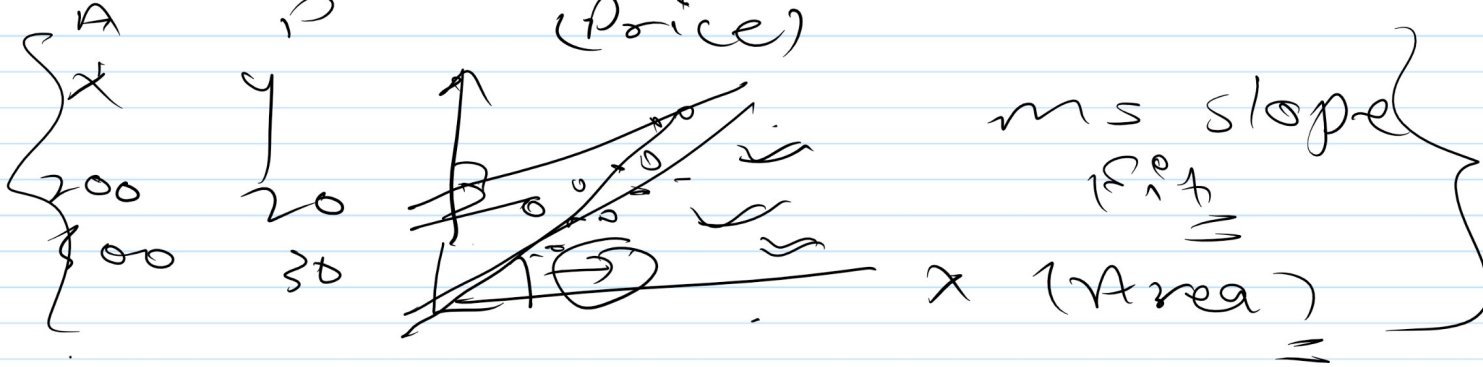
test data -> Model accuracy

validation data -> Hyperparameter tuning

Theory questions with solution -> train -> sample papers solve -> solution (validate)

train
└─┬─ Train
 └─ validation.





- ① Underfitting & overfitting
- ② ~~working~~ Imbalance dataset
- ③ Feature selection & FE